In this homework we will build on top of the work from Homework 5

1. I will assume that you have successfully populated the database with the information coming from the client as described in HW 5. If not please reach out to the TA for a prepopulated database.
2. Convert your database schema to one that conforms to 3rd normal form and write the necessary queries to transfer the data from what you had before to the new schema. If your previous schema is already conforming to 3rd normal form, please explain why this is the case. Hint: We know that a particular IP address is always coming from the same country although we also know that a country has many IPs associated with it.
3. Write a program that can retrieve the data from the database, and build 2 models that use some of the fields to predict some of the other fields. Run this program on a VM that you will create as part of the exercise.
    - The first model should use client IP to predict the country from which the request originated. You can use any kind of model you want but you should be able to achieve at least 99% accuracy for this exercise.
    - The second model should use any of the available fields to predict income. Once again you get to choose what kind of model you want to use. You should aim for 35+% accuracy for this second model but report any problems you run into, as fitting a model for this data is subject to the vagaries of the random seed you may have picked.
    - You should create a test set for each model and show the accuracy of your model on the test set. Write the output of your models on the test sets as separate files in your cloud bucket from the previous homework.

What to turn in:

- A pdf file describing all the necessary steps to configure and run your app, the output of your models and an explanation of how your models work.
- Your pdf file should include your new schema and any necessary queries for transferring data, or the explanation on why your old schema conforms to 3rd normal form.
- Your pdf should contain the location of your bucket with the output of your models
- Your pdf file should include a github link pointing to your python code.
- Your github should include a script for enabling your database, creating a VM and running the code with your 2 models. Make sure your script stops the database when it is done, deletes the VM, and prints the output of the file in your cloud bucket on the terminal.